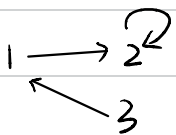
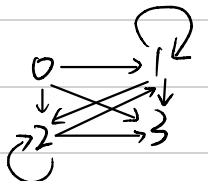
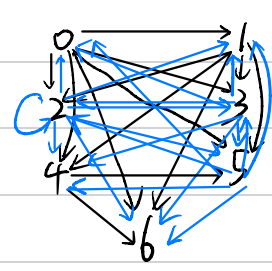


4.1.2 (1)  $M = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(2)  $M = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(3)  $M = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

3. 一元关系: $\sum_{i=0}^n C_n^i = 2^n$

二元关系: $\sum_{i=0}^n C_{n^2}^i = 2^{n^2}$

m元关系: $\sum_{i=0}^{n^m} C_{n^m}^i = 2^{n^m}$

4. 最少: 空关系

最多: $A \times A$

4.2.1 (a) 自反性, 反对称性, 传递性

(b) 反对称性, 传递性

(c) 自反性, 对称性, 传递性

(d) 反自反性, 反对称性,

(e) 反自反性, 反对称性,

(f) 自反性, 对称性, 反对称性, 传递性

2. (1) R_1 : 对称性 R_2 : 对称性, 传递性,

R_3 : 反自反性, 对称性, R_4 : 自反性, 对称性, 传递性,

R_5 : 反对称性

反对称性

(2) R_1 : 对称性 R_2 : 对称性, 传递性,

R_3 : 反自反性, 对称性, R_4 : 自反性, 对称性, 传递性,

R_5 : 反对称性

4. (4) $M = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ 反自反性, 反对称, 传递性

(b) $M = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ 反自反性, 对称性

(c) $M = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 无

	$\cup$	$\cap$	$-$	$A^2 - R_1$	$\tilde{R}_1$	$\circ$
自反	✓	✓	✗	✗	✓	✓
反自反	✓	✓	✓	✗	✓	✗
对称	✓	✓	✓	✓	✓	✗
反对称	✗	✓	✓	✗	✓	✗
传递	✗	✓	✗	✗	✓	✗

1. 自反: (i) $A = \{1, 2, 3\}$,

$R_1 = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 1, 2 \rangle \}$

$R_2 = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle \}$ 自反

$R_1 - R_2 = \{ \langle 1, 2 \rangle \}$ 无自反性

(ii) $A^2 - R_1$, $A = \{1, 2\}$, $R_1 = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle \}$

$A^2 - R_1 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle \}$ 无自反性

2. 反自反: (i) $A^2 - R_1$, $A = \{1, 2\}$, $R_1 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle \}$

有反自反性 $A^2 - R_1 = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle \}$ 无反自反性

(ii) $R_1, R_2: A = \{1, 2\}$, $R_1 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle \}$

$R_2 = \{ \langle 2, 1 \rangle, \langle 1, 2 \rangle \}$, R_1, R_2 均有反自反性

$R_1 \circ R_2 = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle \}$ 无反自反性

3. 对称: (i) $R_1 \circ R_2: A = \{1, 2\}$

$R_1 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle \}$ $R_2 = \{ \langle 1, 1 \rangle \}$

$R_1 \circ R_2 = \{ \langle 2, 1 \rangle \}$,

4. 反对称: (i) $\cup: A = \{1, 2\}$, $R_1 = \{ \langle 1, 2 \rangle \}$, $R_2 = \{ \langle 2, 1 \rangle \}$

$R_1 \cup R_2 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle \}$

(ii) $A^2 - R_1$, $A = \{1, 2\}$, $R_1 = \{ \langle 1, 1 \rangle \}$

$A^2 - R_1 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle, \langle 2, 2 \rangle \}$

(iii) $R_1 \circ R_2: A = \{1, 2, 3\}$, $R_1 = \{ \langle 1, 1 \rangle, \langle 2, 3 \rangle \}$

$$R_2 = \{ \langle 1, 2 \rangle, \langle 3, 1 \rangle \} \quad R_1 \circ R_2 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle \}$$

$$5. \text{传递: } (i) U: A = \{1, 2, 3\}. \quad R_1 = \{ \langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 1, 3 \rangle \}$$

$$R_2 = \{ \langle 2, 1 \rangle \}. \quad R_1 \cup R_2 = \{ \langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 1, 3 \rangle, \langle 2, 1 \rangle \}$$

$$(ii) \because A = \{1, 2\}. \quad R_1 = \{ \langle 1, 2 \rangle, \langle 2, 1 \rangle, \langle 1, 1 \rangle \}$$

$$R_2 = \{ \langle 1, 1 \rangle \}. \quad R_1 \cap R_2 = \{ \langle 1, 1 \rangle, \langle 2, 1 \rangle \}$$

$$(iii) A^2 = R, A = \{1, 2\}. \quad R_1 = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle \}$$

$$A^2 \cap R_1 = \{ \langle 1, 1 \rangle, \langle 2, 1 \rangle \}$$

$$(iv) R_1 \circ R_2 \quad A = \{1, 2, 3\}. \quad R_1 = \{ \langle 1, 1 \rangle, \langle 2, 2 \rangle \}$$

$$R_2 = \{ \langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 1, 3 \rangle \}$$

$$R_1 \circ R_2 = \{ \langle 1, 2 \rangle, \langle 2, 3 \rangle \}$$

$$3. (i) R \circ S = \{ \langle a, c \rangle, \langle a, d \rangle, \langle c, a \rangle, \langle d, a \rangle \}$$

$$R \circ R = \{ \langle a, a \rangle, \langle a, c \rangle, \langle a, b \rangle, \langle a, d \rangle, \langle c, b \rangle \}$$

$$S \circ S = \{ \langle c, c \rangle, \langle c, d \rangle, \langle c, a \rangle \}$$

$$(ii) M_1 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad M_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$M_1 \times M_2 = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad \text{正确}$$

$$M_1^2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{正确}$$

$$M_2^2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{正确}$$

$$4. R, S \text{ 对称}, \therefore \tilde{R} = R, \tilde{S} = S$$

$$\textcircled{1} R \circ S \text{ 对称} \therefore R \circ S = \tilde{R} \circ \tilde{S} = \tilde{S} \circ \tilde{R} = S \circ R$$

$$\textcircled{2} R \circ S = S \circ R = \tilde{S} \circ \tilde{R} = \tilde{R} \circ \tilde{S} \therefore R \circ S \text{ 对称}$$

$$5. (i) \textcircled{1} R \text{ 反自反} \Rightarrow \forall x \in A, \langle x, x \rangle \notin R$$

$$\text{又 } \forall x, y \in A, x \neq y \text{ 时 } \langle x, y \rangle \notin I_A$$

$$\therefore I_A \cap A = \emptyset.$$

$$\textcircled{2} I_A \cap A = \emptyset. \therefore \forall x \in A, \langle x, x \rangle \notin R.$$

$$\therefore R \text{ 反自反}$$

$$(ii) R \text{ 反对称} \Rightarrow \forall \langle x, y \rangle, \langle y, x \rangle \in R \Leftrightarrow x = y$$

$$\therefore \forall \langle x, y \rangle \in R \cap \tilde{R} \Rightarrow \langle x, y \rangle \in R \text{ 且 } \langle x, y \rangle \in \tilde{R}$$

$$\therefore \langle y, x \rangle \in R, \therefore x = y. \therefore \langle x, y \rangle \in I_A.$$

$$\text{故 } R \cap \tilde{R} \subseteq I_A$$

$$\textcircled{2} R \cap \tilde{R} \subseteq I_A. \text{ 假设 } \langle x, y \rangle \in R \text{ 且 } \langle y, x \rangle \in \tilde{R}$$

$$\therefore \langle x, y \rangle \in \tilde{R}. \therefore \langle x, y \rangle \in R \cap \tilde{R}.$$

$$\therefore \langle x, y \rangle \in I_A. \text{ 即 } x = y. \therefore R \text{ 反对称}$$

$$3) \textcircled{1} \forall \langle x, z \rangle \in R \circ R,$$

$$\text{可知 } \exists y \in A, \langle x, y \rangle, \langle y, z \rangle \in R. \text{ 又 } R \text{ 传递}, \langle x, z \rangle \in R.$$

$$\therefore R \circ R \subseteq R$$

$$\textcircled{2} \forall \langle x, z \rangle \in R \circ R, \exists y \in A, \langle x, y \rangle, \langle y, z \rangle \in R.$$

$$\text{又 } R \circ R \subseteq R. \therefore \langle x, z \rangle \in R. \therefore R \text{ 传递}.$$

$$9. (i) R = \{ \langle 0, 1 \rangle, \langle 1, 2 \rangle \}$$

$$(ii) (\Rightarrow): \forall \langle x, z \rangle \in R^2, \exists y \in A, \langle x, y \rangle, \langle y, z \rangle \in R.$$

$$\text{又 } R \text{ 反传递}. \langle x, z \rangle \notin R. \therefore R^2 \cap R = \emptyset$$

$$(\Leftarrow): \forall \langle x, z \rangle \in R^2, \exists y \in A, \langle x, y \rangle, \langle y, z \rangle \in R.$$

$$\text{又 } R^2 \cap R = \emptyset. \therefore \langle x, z \rangle \notin R. \therefore R \text{ 反传递}$$